



**HACETTEPE UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF NUCLEAR ENGINEERING**

NEM 358: Nuclear Thermal Hydraulics
Term Project

Project 3: CFD Simulation of a 3x3 PWR Rod Bundle

Student Name	Student ID	CFD Software
Emre Sakarya	2230386062	ANSYS Fluent
Eren Gürel	2220386044	Star-CCM+

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1 Introduction

In nuclear power plants, fuel bundles constitute the most critical thermal-hydraulic components of the reactor core, as heat transfer begins here. In Pressurized Water Reactors (PWRs), fuel rods are arranged in a square plane at specific intervals, forming sub-channels where coolant enters from the bottom and exits from the top. Accurately predicting the flow and heat transfer behavior in these sub-channels is extremely important for critical and vital issues such as fuel integrity, safety margins, and reactor efficiency.

While traditional analytical methods appear to yield clear results due to simplified geometries and homogeneous flow assumptions, they cannot adequately resolve the local temperature, velocity, and pressure distributions within the sub-channel. Computational Fluid Dynamics (CFD) methods overcome this limitation, offering the capacity to simultaneously resolve the three-dimensional flow field, turbulence effects, and conjugate heat transfer.

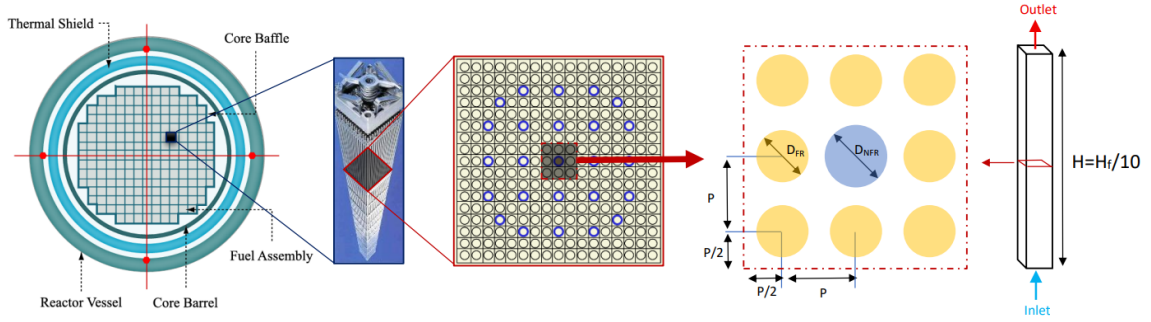


Figure 1: Multi-scale schematic representation of a standard PWR cross-section, showing the fuel assembly configuration, the 3x3 rod bundle matrix with a center non-fuel rod (D_{NFR}), and the modeled axial subchannel domain ($H = H_f/10$).

In this study, a thermal-hydraulic analysis of a 3x3 fuel rod bundle was performed using parameters taken from a typical 4-loop PWR reactor. To reduce computational costs, ANSYS Fluent was used, and a quadrature model (1/4 symmetry) was adopted, taking advantage of the quadrature symmetry of the power distribution. However, the Star-CCM+ solution was based on the entire geometry. These analyses were performed independently using CFD software; the obtained results were validated and analyzed using analytical calculations and comparisons. The main objectives of the study are: to investigate the temperature, velocity, and pressure distributions within the subchannel; to compare CFD results with analytical correlations; and to compare the results of two different CFD software programs internally.

2 Methodology

The thermal-hydraulic verification of the 3x3 Pressurized Water Reactor (PWR) rod bundle is governed by the conservation equations of fluid mechanics coupled with empirical heat transfer and friction factor correlations. The analysis assumes steady-state, one-dimensional flowing liquid under an incompressible fluid model at bulk conditions, with cross-verification provided by the three-dimensional numerical simulations.

2.1 Governing Analytical Formulations

The hydraulic diameter (D_h) for the complex subchannel geometry is defined as four times the flow area divided by the wetted perimeter:

$$D_h = \frac{4 \cdot A_{flow}}{P_{wetted}} \quad (1)$$

The mass flux (G) through the flow channel is evaluated using the mass flow rate ($\dot{m}$) and the cross-sectional flow area (A_{flow}):

$$G = \frac{\dot{m}}{A_{flow}} = \rho \cdot V \quad (2)$$

The dimensionless Reynolds number (Re), which determines the turbulent state of the coolant flow within the subchannel, is given by:

$$Re = \frac{\rho \cdot V \cdot D_h}{\mu} = \frac{G \cdot D_h}{\mu} \quad (3)$$

where ρ is the fluid density, V is the bulk velocity, and μ is the dynamic viscosity evaluated at the bulk average temperature ($T_{avg} = 309.4^\circ\text{C}$).

2.2 Frictional and Hydrostatic Pressure Drops

To account for the pressure drops across the modeled active length segment ($L = 0.366$ m), the Darcy-Weisbach framework is utilized. The Darcy friction factor (f) for fully developed turbulent flow in smooth channels is determined via the Petukhov correlation:

$$f = (0.790 \cdot \ln(Re) - 1.64)^{-2} \quad (4)$$

Under the incompressible flow assumption, fluid acceleration is omitted ($\Delta P_{acceleration} = 0$ Pa). Consequently, the total pressure drop (ΔP_{total}) is a linear combination of the viscous frictional losses ($\Delta P_{friction}$) and the hydrostatic elevation losses due to gravitational forces ($\Delta P_{gravity}$):

$$\Delta P_{total} = \Delta P_{friction} + \Delta P_{gravity} \quad (5)$$

The explicit formulations for these components over the axial length are evaluated as:

$$\Delta P_{friction} = f \cdot \left(\frac{L}{D_h} \right) \cdot \frac{\rho \cdot V^2}{2} \quad (6)$$

$$\Delta P_{gravity} = \rho \cdot g \cdot L \quad (7)$$

where $g = 9.81$ m/s² represents the gravitational acceleration vector acting opposite to the bulk flow direction.

2.3 Boundary Conditions and Solvers

The numerical domain consists of an active height of $H = H_f/10 = 0.366$ m to balance computational efficiency. The boundary conditions applied within the numerical platforms are structured as follows:

- **Inlet Boundary:** Uniform Velocity Inlet based on mass flux requirements at a fixed inlet temperature $T_{in} = 292.7^\circ\text{C}$ (565.85 K).
- **Outlet Boundary:** Pressure Outlet maintained at the reactor nominal operating pressure $P_{nominal} = 15.51$ MPa.
- **Wall Boundary:** Conjugate heat transfer with volumetric heat generation modeling the heat generated via nuclear fission within the active fuel rods.

For the Star-CCM+ model, the full 3x3 geometry was created by constructing a square prism with dimensions $37.8 \text{ mm} \times 37.8 \text{ mm} \times 0.366 \text{ m}$. Eight fuel rods with a diameter of 9.5 mm and one central guide tube with a diameter of 12.243 mm were placed inside the computational domain. Boolean subtraction was applied to separate the coolant volume from the solid rods. This allowed the fluid and solid regions to be solved together under a conjugate heat transfer formulation.

The coolant was modeled as pressurized liquid water at the operating condition. The following constant thermophysical properties were used for the coolant:

Table 1: Coolant material properties used in the CFD model.

Property	Value
Density, ρ	706.2 kg/m ³
Dynamic viscosity, μ	$8.71 \times 10^{-5} \text{ Pa} \cdot \text{s}$
Specific heat, c_p	5950 J/kgK
Thermal conductivity, k	0.54 W/mK
Reference pressure	15.51 MPa

For the solid regions, the fuel rods and the non-fuel guide tube were assigned separate material properties. The fuel rods were treated as internally heated solid regions, while the central guide tube was treated as a non-heated solid region. The Star-CCM+ model used steady-state, three-dimensional, turbulent, segregated flow, segregated fluid energy, and segregated solid energy models. The $k - \omega$ SST turbulence model was selected due to its suitability for wall-bounded turbulent flows and near-wall heat transfer. Gravity was included in the simulation in order to account for the hydrostatic contribution to the total pressure drop.

The imposed radial power distribution was applied using volumetric heat sources. The power fraction matrix was:

$$\begin{bmatrix} 0.85 & 1.00 & 0.85 \\ 1.00 & 0 & 1.00 \\ 0.85 & 1.00 & 0.85 \end{bmatrix} \quad (8)$$

The rods with a power fraction of 1.00 were assigned:

$$q'''_{1.00} = 2.5136 \times 10^8 \text{ W/m}^3 \quad (9)$$

The corner rods with a power fraction of 0.85 were assigned:

$$q'''_{0.85} = 2.1366 \times 10^8 \text{ W/m}^3 \quad (10)$$

The central guide tube had no heat generation:

$$q'''_{guide} = 0 \text{ W/m}^3 \quad (11)$$

The Star-CCM+ mesh was generated using an automated polyhedral meshing strategy. Prism layers were applied near the rod surfaces to resolve near-wall gradients in temperature and velocity. The final Star-CCM+ mesh contained approximately 2.11×10^6 cells.

3 Results and Discussion

The obtained numerical results were evaluated using three main thermal-hydraulic quantities: temperature, pressure, and velocity. These quantities were investigated separately for ANSYS Fluent and Star-CCM+ simulations and were compared with the analytical calculations derived in the previous section. The ANSYS Fluent model was solved using a reduced symmetry-based computational domain, whereas the Star-CCM+ model was solved using the full 3x3 rod bundle geometry. Therefore, small differences between the two solvers are expected due to differences in geometrical representation, mesh structure, near-wall treatment, and numerical discretization.

The individual task distribution within the project was organized according to the CFD software used. The ANSYS Fluent setup, solution, and post-processing were performed by Emre Sakarya, while the Star-CCM+ setup, solution, post-processing, and analytical comparison were performed by Eren Gürel. The analytical calculations were used as a common benchmark for the interpretation of both CFD solutions.

3.1 Analytical Benchmark Results

Before evaluating the CFD results, analytical reference values were calculated to provide a benchmark for the expected thermal-hydraulic behavior. The coolant flow area was calculated as:

$$A_{flow} = 7.4406 \times 10^{-4} \text{ m}^2 \quad (12)$$

Using the average core mass flux of $G = 3729 \text{ kg/m}^2\text{s}$, the mass flow rate through the 3x3 model was obtained as:

$$\dot{m} = GA_{flow} \quad (13)$$

$$\dot{m} = 3729 \times 7.4406 \times 10^{-4} \quad (14)$$

$$\dot{m} = 2.775 \text{ kg/s} \quad (15)$$

The inlet velocity corresponding to this mass flux was:

$$V = \frac{G}{\rho} = \frac{3729}{706.2} \quad (16)$$

$$V = 5.28 \text{ m/s} \quad (17)$$

The hydraulic diameter of the subchannel was calculated as:

$$D_h = 0.01074 \text{ m} \quad (18)$$

The Reynolds number was then obtained as:

$$Re = \frac{\rho V D_h}{\mu} \quad (19)$$

$$Re = \frac{706.2 \times 5.28 \times 0.01074}{8.71 \times 10^{-5}} \quad (20)$$

$$Re = 4.60 \times 10^5 \quad (21)$$

Since the Reynolds number is much larger than 4000, the coolant flow is fully turbulent. Therefore, the turbulent flow assumption used in both CFD solvers is justified.

The total heat input of the 3x3 model was determined by applying the given radial power distribution. The total power fraction of the 3x3 matrix is:

$$4(0.85) + 4(1.00) + 0 = 7.4 \quad (22)$$

The average heat generation rate assigned to a fuel rod segment with power fraction 1.00 was:

$$q'''_{1.00} = 2.5136 \times 10^8 \text{ W/m}^3 \quad (23)$$

For the corner rods with a power fraction of 0.85:

$$q'''_{0.85} = 2.1366 \times 10^8 \text{ W/m}^3 \quad (24)$$

The total heat input to the 3x3 model was:

$$\dot{Q}_{total} = 48251.54 \text{ W} \quad (25)$$

The analytical coolant temperature rise was calculated from the steady-flow energy balance:

$$\Delta T = \frac{\dot{Q}_{total}}{\dot{m}c_p} \quad (26)$$

Using $c_p = 5950 \text{ J/kgK}$:

$$\Delta T = \frac{48251.54}{2.775 \times 5950} \quad (27)$$

$$\Delta T = 2.92 \text{ K} \quad (28)$$

Thus, the analytically estimated average coolant outlet temperature is:

$$T_{out,avg} = T_{in} + \Delta T \quad (29)$$

$$T_{out,avg} = 565.85 + 2.92 = 568.77 \text{ K} \quad (30)$$

For the pressure drop calculation, the frictional and hydrostatic components were evaluated separately:

$$\Delta P_{friction} = f \left(\frac{L}{D_h} \right) \frac{\rho V^2}{2} \quad (31)$$

$$\Delta P_{gravity} = \rho g L \quad (32)$$

The frictional pressure drop was calculated as approximately:

$$\Delta P_{friction} = 4548.86 \text{ Pa} \quad (33)$$

The hydrostatic pressure drop was:

$$\Delta P_{gravity} = 706.2 \times 9.81 \times 0.366 \quad (34)$$

$$\Delta P_{gravity} = 2535.58 \text{ Pa} \quad (35)$$

Therefore, the analytical total pressure drop was:

$$\Delta P_{total} = 4548.86 + 2535.58 \quad (36)$$

$$\Delta P_{total} = 7084.44 \text{ Pa} \quad (37)$$

$$\Delta P_{total} = 7.08 \text{ kPa} \quad (38)$$

Table 2: Summary of analytical benchmark values.

Parameter	Value
Flow area, A_{flow}	$7.4406 \times 10^{-4} \text{ m}^2$
Hydraulic diameter, D_h	0.01074 m
Mass flow rate, $\dot{m}$	2.775 kg/s
Inlet velocity, V	5.28 m/s
Reynolds number, Re	4.60×10^5
Total heat input, $\dot{Q}_{total}$	48.25 kW
Analytical coolant temperature rise	2.92 K
Analytical outlet average temperature	568.77 K
Analytical total pressure drop	7.08 kPa

3.2 ANSYS Fluent Results

The ANSYS Fluent solution was post-processed using outlet contour plots, diagonal line profiles, and an axial temperature line plot. The outlet temperature contour is shown in Fig. 2. The temperature field clearly follows the imposed radial power distribution. The regions corresponding to heated fuel rods exhibit higher temperatures, while the coolant region remains close to the inlet coolant temperature. Since the plotted contour includes the heated solid regions, the maximum temperature in the plot represents the solid fuel rod temperature rather than the bulk coolant outlet temperature.

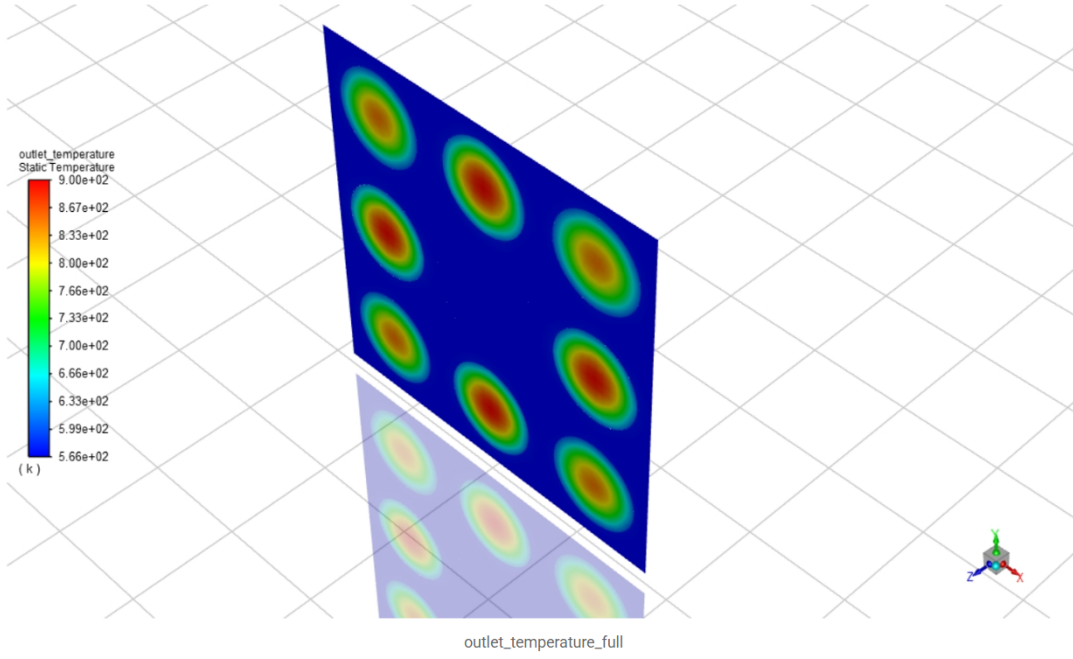


Figure 2: ANSYS Fluent outlet temperature contour for the 3x3 rod bundle. The high-temperature regions correspond to heated fuel rods, while the coolant region remains close to the inlet coolant temperature.

The diagonal temperature profile at the outlet is shown in Fig. 3. The profile contains sharp temperature increases when the diagonal line crosses heated rod regions. The highest temperature along this line is approximately 850 K, whereas the coolant portions of the line remain close to 566–570 K. This behavior is physically consistent because volumetric heat generation occurs inside the solid fuel rods, and the coolant temperature is much lower than the internal fuel temperature.

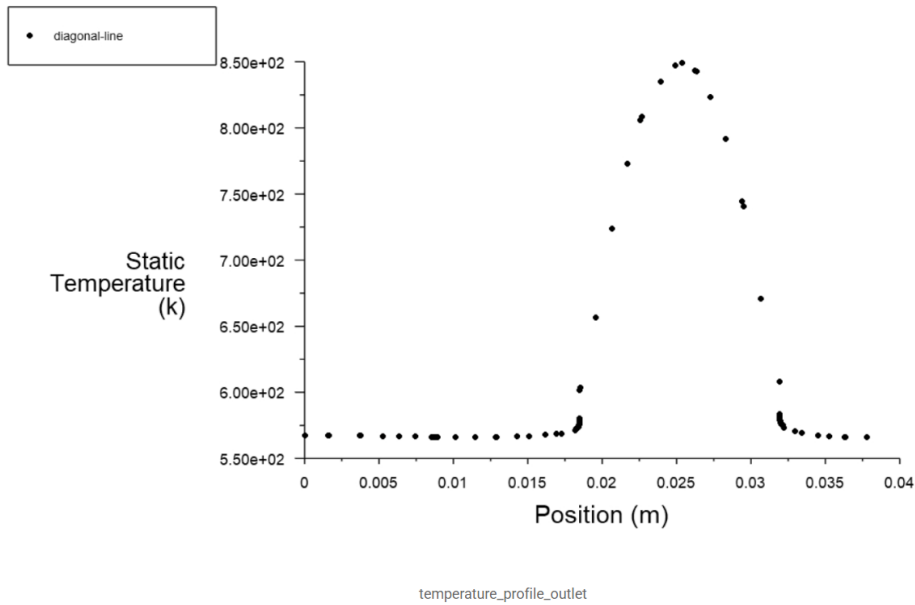


Figure 3: ANSYS Fluent diagonal temperature profile at the outlet plane. The peaks correspond to the heated solid rod regions crossed by the diagonal line.

The axial temperature profile is presented in Fig. 4. The temperature increases gradually along the flow direction from the inlet to the outlet. This confirms that energy is transferred from the fuel rods to the coolant as the water flows upward through the rod bundle. Although the local axial line temperature increase is small, the trend is monotonic and agrees with the expected heat addition mechanism.

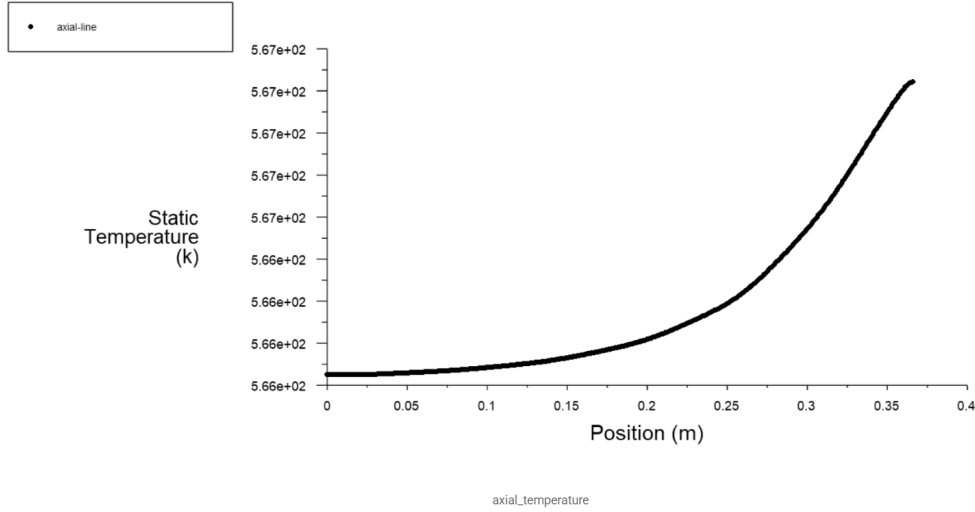


Figure 4: ANSYS Fluent axial temperature profile. The coolant temperature increases along the axial direction due to heat transfer from the fuel rods.

The outlet pressure contour obtained from ANSYS Fluent is shown in Fig. 5. Since the outlet was defined as a pressure outlet, the gauge pressure is expected to be close to zero at this boundary. The contour indicates small local variations near the walls and narrow flow passages, which are associated with local velocity gradients and wall effects. However, the total pressure drop of the channel must be evaluated between the inlet and outlet rather than only from the outlet plane contour.

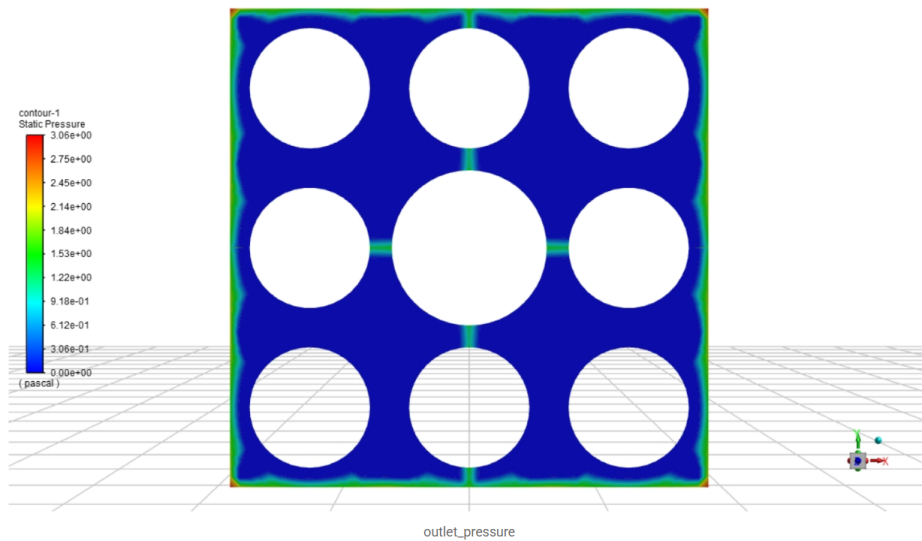


Figure 5: ANSYS Fluent outlet static pressure contour. Since the outlet is defined as a pressure outlet, the static gauge pressure remains close to zero over most of the outlet plane.

The diagonal pressure profile is shown in Fig. 6. Similar to the outlet pressure contour, the static pressure remains close to zero over most of the outlet plane. Small pressure peaks appear where the line passes close to wall or narrow-gap regions. These local variations are acceptable because the outlet boundary condition fixes the reference pressure at the exit.

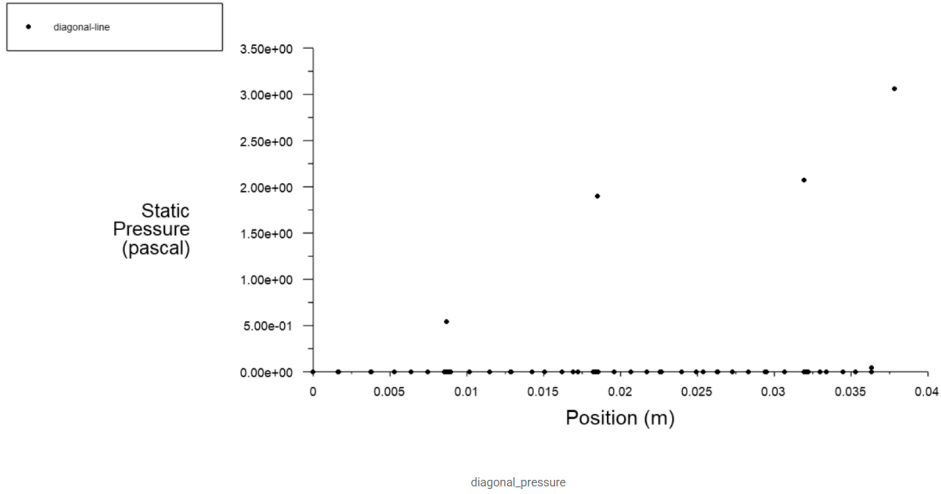


Figure 6: ANSYS Fluent diagonal static pressure profile at the outlet plane. Local peaks are observed near narrow flow passages and wall-adjacent regions.

The outlet velocity contour is shown in Fig. 7. The velocity magnitude increases in the narrow subchannels between rods. This is expected because the coolant must pass through reduced flow areas between the solid rods. The maximum velocity in the ANSYS Fluent result reaches approximately 6.25 m/s, which is higher than the inlet bulk velocity of 5.28 m/s due to local flow acceleration.

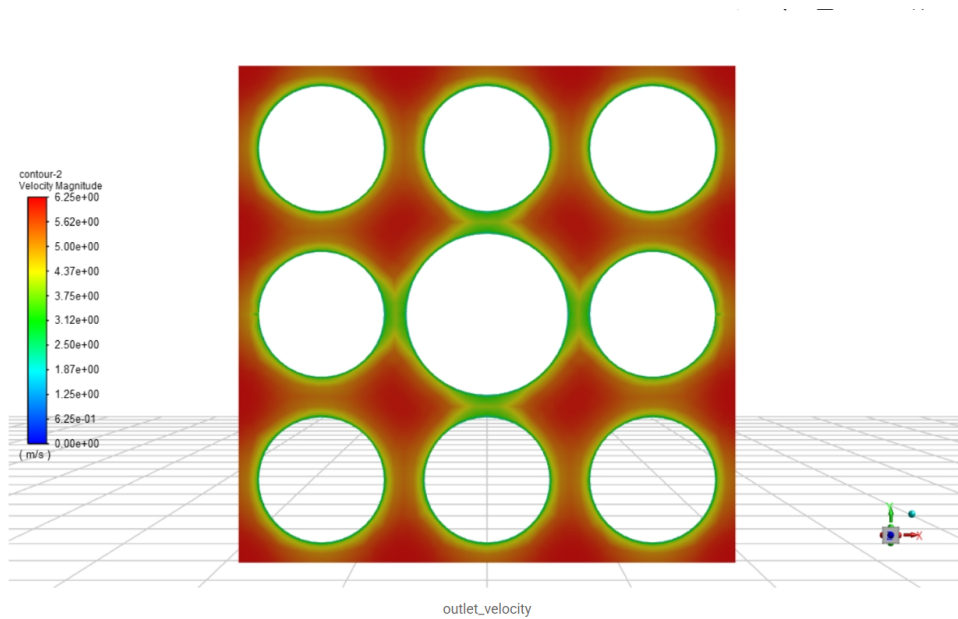


Figure 7: ANSYS Fluent outlet velocity magnitude contour. Higher velocities are observed in the narrow flow passages between rods.

The diagonal velocity profile is shown in Fig. 8. The profile contains zero-velocity regions where the diagonal line crosses solid rod regions and high-velocity peaks in the coolant subchannels. This confirms the presence of no-slip solid boundaries and local acceleration in the open coolant passages.

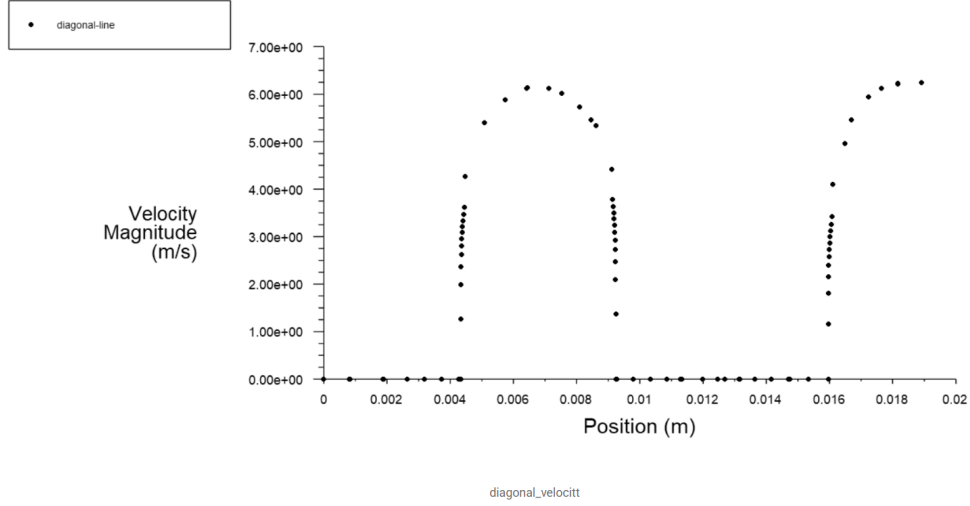


Figure 8: ANSYS Fluent diagonal velocity profile at the outlet plane. Zero values correspond to solid regions, while velocity peaks correspond to coolant passages.

The residual history is shown in Fig. 9. The continuity, momentum, energy, k , and ω residuals decrease significantly during the solution process. The energy residual reaches a very low order, while the remaining equations also show stable convergence. Therefore, the solution can be considered numerically stable for the purpose of thermal-hydraulic comparison.

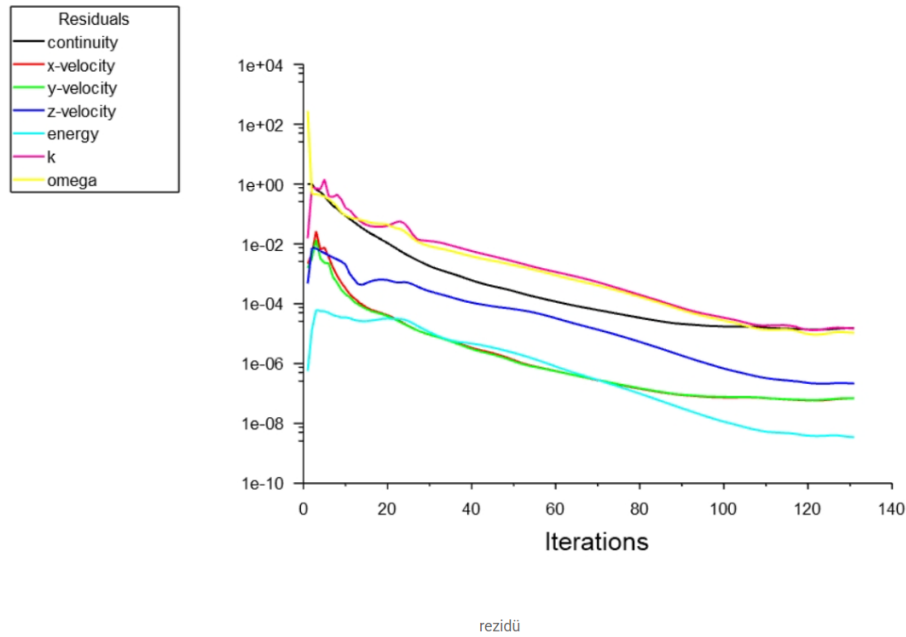


Figure 9: ANSYS Fluent residual history. The residuals decrease steadily, indicating a converged steady-state numerical solution.

The total pressure drop calculated by ANSYS Fluent was:

$$\Delta P_{ANSYS} = 6953 \text{ Pa} \quad (39)$$

$$\Delta P_{ANSYS} = 6.953 \text{ kPa} \quad (40)$$

This value is close to the analytical pressure drop of 7.08 kPa. The percentage difference is:

$$\% \text{ Difference}_{ANSYS} = \frac{|\Delta P_{analytical} - \Delta P_{ANSYS}|}{\Delta P_{analytical}} \times 100 \quad (41)$$

$$\% \text{ Difference}_{ANSYS} = \frac{|7.08 - 6.953|}{7.08} \times 100 \quad (42)$$

$$\% \text{ Difference}_{ANSYS} = 1.79\% \quad (43)$$

3.3 Star-CCM+ Results

The Star-CCM+ model was solved using the full 3x3 geometry. Therefore, unlike the ANSYS Fluent symmetry model, the complete rod bundle cross-section was directly represented. The outlet velocity contour obtained from Star-CCM+ is presented in Fig. 10. Similar to the ANSYS Fluent result, the coolant accelerates in the narrow gaps between the rods. The maximum velocity reaches approximately 5.92 m/s, while lower velocities are observed near the rod walls due to the no-slip boundary condition.

ter STAR-CCM+

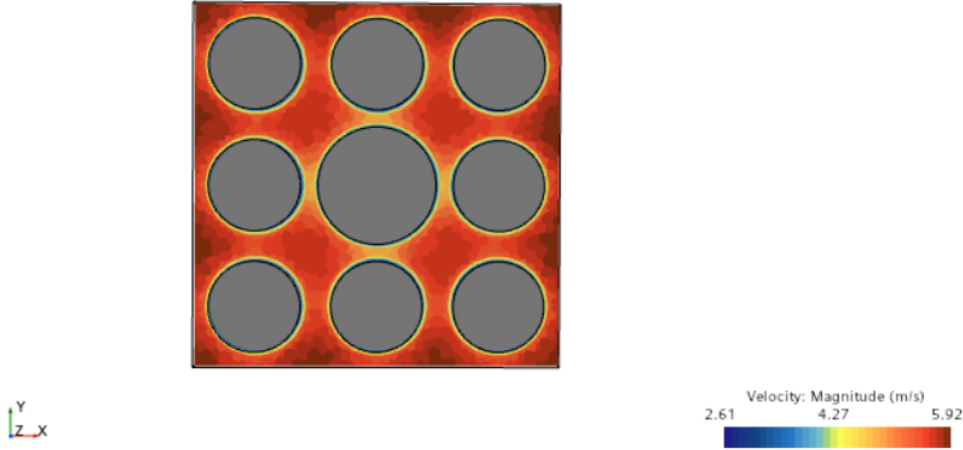


Figure 10: Star-CCM+ outlet velocity magnitude contour for the full 3x3 rod bundle. The coolant accelerates in the narrow subchannels between rods.

The longitudinal temperature distribution obtained from Star-CCM+ is shown in Fig. 11. This side-sectional view clearly shows that the heated fuel rods have high temperatures along the active length, while the coolant region remains much closer to the inlet temperature. The central non-fuel guide tube remains significantly colder because no volumetric heat source was assigned to it. This temperature difference is physically expected due to the internal volumetric heat generation in the fuel rods and the finite thermal conductivity of the solid material.



Figure 11: Star-CCM+ side-sectional temperature distribution. Heated fuel rods show high temperatures along the active length, while the coolant and the central non-fuel guide tube remain cooler.

The outlet temperature distribution obtained from Star-CCM+ is shown in Fig. 12. The result demonstrates the expected radial power pattern. The rods with a power fraction of 1.00 reach higher temperatures than the corner rods with a power fraction of 0.85. The central guide tube has no heat generation and therefore remains at a much lower temperature. The maximum solid fuel temperature at the outlet section is approximately 938 K. This maximum temperature belongs to the heated solid region and should not be interpreted as the coolant outlet temperature. The coolant temperature is much lower and is better represented by mass-flow-averaged outlet temperature values or coolant-only profiles.

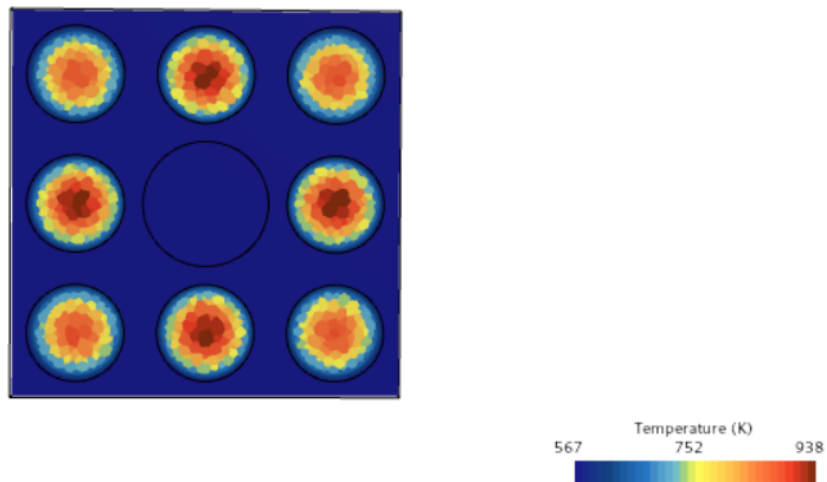


Figure 12: Star-CCM+ outlet temperature contour showing the radial power distribution. The rods with higher power fraction exhibit higher temperatures, while the central non-fuel guide tube remains cooler.

The total pressure drop calculated by Star-CCM+ between the inlet and outlet was:

$$\Delta P_{StarCCM+} = 6494 \text{ Pa} \quad (44)$$

$$\Delta P_{StarCCM+} = 6.494 \text{ kPa} \quad (45)$$

This value includes both the wall-friction contribution and the hydrostatic contribution due to gravity. The analytical total pressure drop was calculated as 7.08 kPa, which is close to the Star-CCM+ result. The percentage difference is:

$$\% \text{ Difference}_{StarCCM+} = \frac{|\Delta P_{analytical} - \Delta P_{StarCCM+}|}{\Delta P_{analytical}} \times 100 \quad (46)$$

$$\% \text{ Difference}_{StarCCM+} = \frac{|7.08 - 6.494|}{7.08} \times 100 \quad (47)$$

$$\% \text{ Difference}_{StarCCM+} = 8.28\% \quad (48)$$

This difference is acceptable considering the simplifications used in the analytical calculation. The analytical model assumes a one-dimensional bulk flow, an empirical friction factor, and uniform average properties, whereas the CFD model resolves the three-dimensional velocity field, near-wall gradients, and local subchannel effects.

3.4 Comparison of ANSYS Fluent, Star-CCM+, and Analytical Results

The main results obtained from analytical calculations, ANSYS Fluent, and Star-CCM+ are summarized in Table 3. Both CFD solvers show the same physical trends: the coolant accelerates in the narrow subchannels, the heated rods reach higher temperatures than the coolant, and the non-fuel center rod remains cooler due to the absence of internal heat generation.

Table 3: Comparison of analytical, ANSYS Fluent, and Star-CCM+ results.

Quantity	Analytical	ANSYS Fluent	Star-CCM+
Inlet velocity	5.28 m/s	5.28 m/s	5.28 m/s
Reynolds number	4.60×10^5	Turbulent	Turbulent
Coolant temperature rise	2.92 K	Monotonic increase	Monotonic increase
Maximum outlet velocity	–	$\approx 6.25 \text{ m/s}$	$\approx 5.92 \text{ m/s}$
Maximum solid temperature	–	$\approx 900 \text{ K}$	$\approx 938 \text{ K}$
Total pressure drop	7.08 kPa	6.953 kPa	6.494 kPa
Pressure drop difference	–	1.79%	8.28%

The pressure drop comparison indicates that both CFD solvers are in acceptable agreement with the analytical benchmark. ANSYS Fluent gives a pressure drop of 6.953 kPa, which differs from the analytical value by only 1.79%. Star-CCM+ gives a pressure drop of 6.494 kPa, with a difference of 8.28%. The difference between the two CFD solvers can be attributed to differences in computational domain representation, mesh topology, wall treatment, discretization schemes, and the fact that ANSYS Fluent used a reduced symmetry model while Star-CCM+ used the full 3x3 geometry.

The velocity results show similar behavior in both CFD solvers. ANSYS Fluent predicts a maximum outlet velocity of approximately 6.25 m/s, while Star-CCM+ predicts approximately 5.92 m/s. In both cases, the maximum velocity occurs in the narrow coolant passages between rods. This is physically expected because the coolant accelerates locally when the available flow area decreases.

The temperature results also show consistent thermal behavior. In both solvers, the heated rods reach much higher temperatures than the coolant because heat is generated volumetrically inside the solid fuel region. The coolant temperature remains close to the inlet temperature, with a small average temperature increase predicted analytically. This is reasonable because the analyzed active length is only one tenth of the full fuel height. The outlet temperature contours also confirm the radial power distribution: rods with a power fraction of 1.00 are hotter than rods with a power fraction of 0.85, while the central non-fuel guide tube remains colder.

Overall, the analytical calculation, ANSYS Fluent result, and Star-CCM+ result are consistent with each other. The pressure drop agreement is especially strong for the ANSYS Fluent result and acceptable for the Star-CCM+ result. The velocity and temperature fields from both CFD solvers also confirm the expected thermal-hydraulic behavior of a PWR rod bundle.

4 Conclusions

In this project, a thermal-hydraulic analysis of a reduced 3x3 PWR rod bundle was performed using ANSYS Fluent and Star-CCM+. The main purpose was to investigate the temperature, pressure, and velocity distributions inside a representative rod bundle subchannel and to compare CFD results with analytical calculations.

The geometry was constructed using the given rod pitch, fuel rod diameter, guide tube diameter, and reduced active height of $H = H_f/10 = 0.366$ m. Eight rods were modeled as heated fuel rods, while the center rod was modeled as a non-fuel guide tube with zero volumetric heat generation. The imposed radial power distribution was successfully represented by assigning different volumetric heat sources to rods with power fractions of 1.00 and 0.85.

The analytical calculations showed that the coolant flow was fully turbulent, with a Reynolds number of approximately 4.60×10^5 . The analytical coolant temperature rise over the reduced active length was calculated as approximately 2.92 K, and the analytical total pressure drop including both frictional and hydrostatic effects was calculated as 7.08 kPa.

The ANSYS Fluent results showed physically consistent temperature, pressure, and velocity distributions. The outlet temperature contour and diagonal temperature profile demonstrated that the heated fuel rods reached much higher temperatures than the coolant. The outlet velocity contour showed acceleration in narrow subchannels, with a maximum velocity of approximately 6.25 m/s. The total pressure drop was obtained as 6.953 kPa, which differs from the analytical value by only 1.79%. This close agreement confirms that the ANSYS Fluent result is consistent with the analytical pressure drop estimation.

The Star-CCM+ results also showed the expected thermal-hydraulic behavior. The full 3x3 geometry clearly demonstrated the radial temperature distribution caused by the imposed power peaking matrix. The side-sectional temperature distribution showed high temperatures along the heated rods and a cooler central guide tube. The outlet

temperature contour showed that the rods with a power fraction of 1.00 reached higher temperatures than the corner rods with a power fraction of 0.85. The maximum solid fuel temperature reached approximately 938 K, while the central non-fuel rod remained cooler. The maximum outlet velocity was approximately 5.92 m/s. The calculated total pressure drop was 6.494 kPa, which differed from the analytical value by approximately 8.28%.

Overall, both CFD solvers produced results that are consistent with the analytical calculations and with the expected physical behavior of a PWR rod bundle. The coolant accelerated in the narrow gaps between rods, the fuel rods exhibited higher temperatures due to internal volumetric heat generation, and the pressure drop was governed by the combined effects of wall friction and gravity. Therefore, the reduced 3x3 model can be considered an appropriate engineering simplification for investigating the local thermal-hydraulic behavior of a PWR rod bundle.

References

- [1] Todreas, N. E., & Kazimi, M. S. (2011). *Nuclear Systems Volume I: Thermal Hydraulic Fundamentals*. CRC Press.

Appendix

This appendix summarizes the main numerical values used in the analytical calculations and CFD setup.

Table 4: Geometrical parameters of the 3x3 rod bundle model.

Parameter	Value
Rod pitch, P	12.6 mm
Fuel rod diameter, D_f	9.5 mm
Guide tube diameter, D_g	12.243 mm
Active fuel height, H_f	3.66 m
Model height, $H_f/10$	0.366 m
Domain cross-section	37.8 mm $\times$ 37.8 mm

Table 5: Main boundary conditions used in the CFD simulations.

Boundary / Quantity	Value
Inlet velocity	5.28 m/s
Inlet temperature	565.85 K
Outlet gauge pressure	0 Pa
Reference pressure	15.51 MPa
Gravity	Included
Flow regime	Turbulent
Turbulence model	$k - \omega$ SST

Table 6: Volumetric heat source values used in the fuel rods.

Rod Type	Volumetric Heat Source
Fuel rods with power fraction 1.00	$2.5136 \times 10^8 \text{ W/m}^3$
Fuel rods with power fraction 0.85	$2.1366 \times 10^8 \text{ W/m}^3$
Central guide tube	0 W/m^3